

主观题 HW9

对于本次作业中注明了“写出”的题目：你可以给出简要计算过程，也可以直接写出结果。如果你写出了计算过程，那么若你的结果不正确，也可能根据过程得到部分分数；如果你直接写出结果，那么或者得满分，或者得零分。

HW 9.1 (2×5 分)

写出下列集合：

$$(1) \cup\{PPP(\emptyset), PP(\emptyset), P(\emptyset), \emptyset\}$$

$$(2) \cap\{PPP(\emptyset), PP(\emptyset), P(\emptyset)\}$$

其中：

$$\begin{aligned} (1) \quad P(\emptyset) &= \{\emptyset\} & PP(A) &= P(P(A)) \\ & & PPP(A) &= P(P(P(A))) \\ P(P(\emptyset)) &= P(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\} \\ P(P(P(\emptyset))) &= \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\} \\ \cup\{PPP(\emptyset), PP(\emptyset), P(\emptyset), \emptyset\} &= \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\} \\ (2) \quad \cap\{PPP(\emptyset), PP(\emptyset), P(\emptyset)\} &= \{\emptyset\} \end{aligned}$$

HW 9.2 (2×5 分)

设 $A = \{\{\emptyset\}, \{\{\emptyset\}\}\}$ ，写出集合：

$$(1) P(A) \text{ 和 } \cup P(A)$$

$$(2) \cup A \text{ 和 } P(\cup A)$$

$$\begin{aligned} (1) \quad P(A) &= \{\emptyset, \{\{\emptyset\}\}, \{\{\{\emptyset\}\}\}, \{\{\emptyset\}, \{\{\emptyset\}\}\}\} \\ \cup P(A) &= \{\{\emptyset\}, \{\{\emptyset\}\}\} \\ (2) \quad \cup A &= \{\emptyset, \{\emptyset\}\} \\ P(\cup A) &= \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\} \end{aligned}$$

HW 9.3 (3×5 分)

设 A, B, C 是任意的集合, 证明:

$$(1) (A - B) - C = (A - C) - (B - C)$$

$$(2) A = B \Leftrightarrow A \oplus B = \emptyset$$

$$(3) A \cap B = \emptyset \Leftrightarrow A \subseteq -B \Leftrightarrow B \subseteq -A$$

$$\begin{aligned} (1) & (A - C) - (B - C) \\ &= (A \cap -C) \cap -(B \cap -C) \\ &= (A \cap -C) \cap (B \cup C) \\ &= (A \cap -B \cap -C) \cup C \\ &= (A \cup C) \cap (-B \cup C) \cap E \\ &= (A \cup C) \cap (-B \cup C) \\ &= (A \cap -B) \cup C \\ &= (A - B) - C \end{aligned}$$

不对, 重做

$$(2) A = B \Rightarrow A \oplus B = (A - B) \cup (B - A) = \emptyset \cup \emptyset = \emptyset$$

$$\begin{aligned} A \oplus B = \emptyset & \Leftrightarrow (A - B) \cup (B - A) = \emptyset \\ & \Leftrightarrow A - B = \emptyset \wedge B - A = \emptyset \Leftrightarrow A = B \end{aligned}$$

$$(3) \textcircled{1} \text{ 若 } A \cap B = \emptyset$$

假设 $A \not\subseteq -B$

$$\text{即 } \exists x, x \in A \wedge x \notin -B$$

$$\Leftrightarrow \exists x, x \in A \wedge x \in B$$

$$\Leftrightarrow A \cap B \neq \emptyset \text{ 矛盾}$$

$$\therefore A \subseteq -B$$

$$\textcircled{2} \text{ 若 } A \subseteq -B$$

$$\text{假设 } B \not\subseteq -A \text{ 即 } \exists x, x \in B \wedge x \notin -A$$

$$\Leftrightarrow \exists x, x \in B \wedge x \in A \Leftrightarrow \exists x, x \notin -B \wedge x \in A$$

$$\Leftrightarrow A \not\subseteq -B \text{ 矛盾 } \therefore B \subseteq -A$$

$$\textcircled{3} \text{ 若 } B \subseteq -A \text{ 则 } A \cap B \neq \emptyset$$

$$\Leftrightarrow \exists x, x \in A \wedge x \in B \Leftrightarrow \exists x, x \in B \wedge x \notin -A$$

$$\Leftrightarrow B \not\subseteq -A \text{ 矛盾 } \therefore A \cap B = \emptyset$$

HW 9.4 (4×2.5 分)

说明满足下列条件的集合 A 和 B 有什么关系：

$$(1) A - B = B$$

$$(2) A - B = B - A$$

$$(3) A \cap B = A \cup B$$

$$(4) A \oplus B = A$$

要求给出尽可能全面且简洁的关系式 (如 $A \subseteq B$ 、 $A = B$ 、 $A = \emptyset$ 等)。

$$\textcircled{1}) A - B = A \cap -B = B$$

$$\text{若 } B \neq \emptyset \text{ 则 } \forall x,$$

$$x \in B \Leftrightarrow x \in (A \cap -B)$$

$$\Rightarrow x \in -B \text{ 矛盾}$$

$$\therefore B = \emptyset$$

$$A - B = B$$

$$A - \emptyset = \emptyset \quad A = \emptyset$$

$$\therefore A = B = \emptyset$$

$$\textcircled{2}) A \cap -B = B \cap -A$$

$$-A \cup B = -B \cup A$$

$$\text{若 } A \neq B$$

$$\text{即 } \exists x, x \in A \wedge x \notin B$$

$$\Leftrightarrow x \in A \wedge x \in -B$$

$$\Leftrightarrow x \in (A \cap -B)$$

$$\Leftrightarrow x \in (B \cap -A)$$

$$\Leftrightarrow x \in B \wedge x \in -A$$

$$\Leftrightarrow x \in B \wedge x \notin A$$

$$\text{矛盾 } \therefore A = B$$

(3) 假设 $A \neq B \quad \exists x, x \in A \wedge x \notin B$
 $\Leftrightarrow \exists x, x \notin (A \cap B)$
 但 $\because x \in A \Rightarrow x \in (A \cup B)$

$\therefore \exists x, x \in (A \cap B) \wedge x \in (A \cup B)$, 矛盾
 $\therefore A = B$

HW 9.5 (2×10 分)

给出下列命题成立的充要条件：

(1) $(A - B) \cup (A - C) = A$

(2) $(A - B) \oplus (A - C) = \emptyset$

要求化至尽可能简单的形式。

1) $\forall x, x \in A \Rightarrow x \in (A - B) \vee x \in (A - C)$
 $\Rightarrow x \in (A \cap -B) \vee x \in (A \cap -C)$
 $\Rightarrow x \in -B \vee x \in -C$
 $\Rightarrow x \in (-B \cup -C)$
 $\Rightarrow x \notin (B \cap C)$

$\therefore A \cap B \cap C = \emptyset$

若 $A \cap B \cap C = \emptyset$

则 $\forall x, x \in A \Rightarrow x \notin (B \cap C)$

$\Rightarrow x \in (-B \cup -C)$

$\Rightarrow x \in (A \cap -B) \cup (A \cap -C)$

$\Rightarrow x \in (A - B) \cup (A - C) \quad \therefore A \subseteq (A - B) \cup (A - C)$

(4) $(A - B) \cup (B - A) = A$

假设 $B \neq \emptyset$.

假设 $B \subseteq A$

则 $B \cap -A = \emptyset$

$(A - B) \cup (B - A)$

$= (A - B) \cup \emptyset$

$= A - B \subseteq A$ 矛盾

假设 $B \not\subseteq A$

则 $\exists x, x \in B \wedge x \notin A$

$\Leftrightarrow \exists x, x \in B \wedge x \in -A$

$\Leftrightarrow \exists x, x \in (B - A)$

$\Rightarrow \exists x, x \in (B - A) \cup (A - B)$

但 $x \notin A$ 与 $(A - B) \cup (B - A) = A$ 矛盾

综上 $B = \emptyset$

$$\forall x, x \in (A-B) \cup (A-C) \Rightarrow x \in (A \cap -B) \cup (A \cap -C)$$

$$\Rightarrow x \in A \cap (-B \cup -C) \Rightarrow x \in A \cap -(B \cap C)$$

$$\Rightarrow x \in A \quad \therefore (A-B) \cup (A-C) \subseteq A$$

$$\therefore (A-B) \cup (A-C) = A$$

$$\therefore \text{充要条件: } A \cap B \cap C = \emptyset$$

$$(2) (A-B) \oplus (A-C) = \emptyset$$

$$(A-B) - (A-C) \cup (A-C) - (A-B) = \emptyset$$

$$\Leftrightarrow (A-B) - (A-C) = \emptyset \wedge (A-C) - (A-B) = \emptyset$$

$$\Leftrightarrow A-B = A-C$$

(1) 若 $A \times B = \emptyset$, 则 A 和 B 应满足什么条件?

$$\{ \langle x, y \rangle, x \in A \wedge y \in B \} = \emptyset$$

$$\Rightarrow A = \emptyset \wedge B = \emptyset$$

(2) 在 (1) 的基础上, 对于集合 A , 是否可能满足 $A = A \times A$? 若满足, 给出例子; 若不满足, 说明原因。

$$A = \emptyset \text{ 时}$$

$$A \times A = \emptyset$$

$$A = A \times A$$